RUI CODEX

Causeway

The agent that argues with itself before it tells you why.

Zetaris | Meterless · Nightshift | NVIDIA Nemotron | Cursor

THE PROBLEM

“Why did our margin fall?” is asked every quarter, and answered with a guess.

Who asks

Owner-managers of small Belgian businesses, such as a chocolatier, a café or a bike shop (the demo businesses are synthetic).

The accountants who advise them and have to explain the numbers.

Why it is hard

The ledger lives in one system; the drivers that might explain it (cocoa, energy, tourism, retail volume, bankruptcies) live at Eurostat.

A chatbot writes a fluent paragraph of “possible reasons”. Nobody can check its numbers, and rival explanations are never tested.

WHAT CAUSEWAY DOES

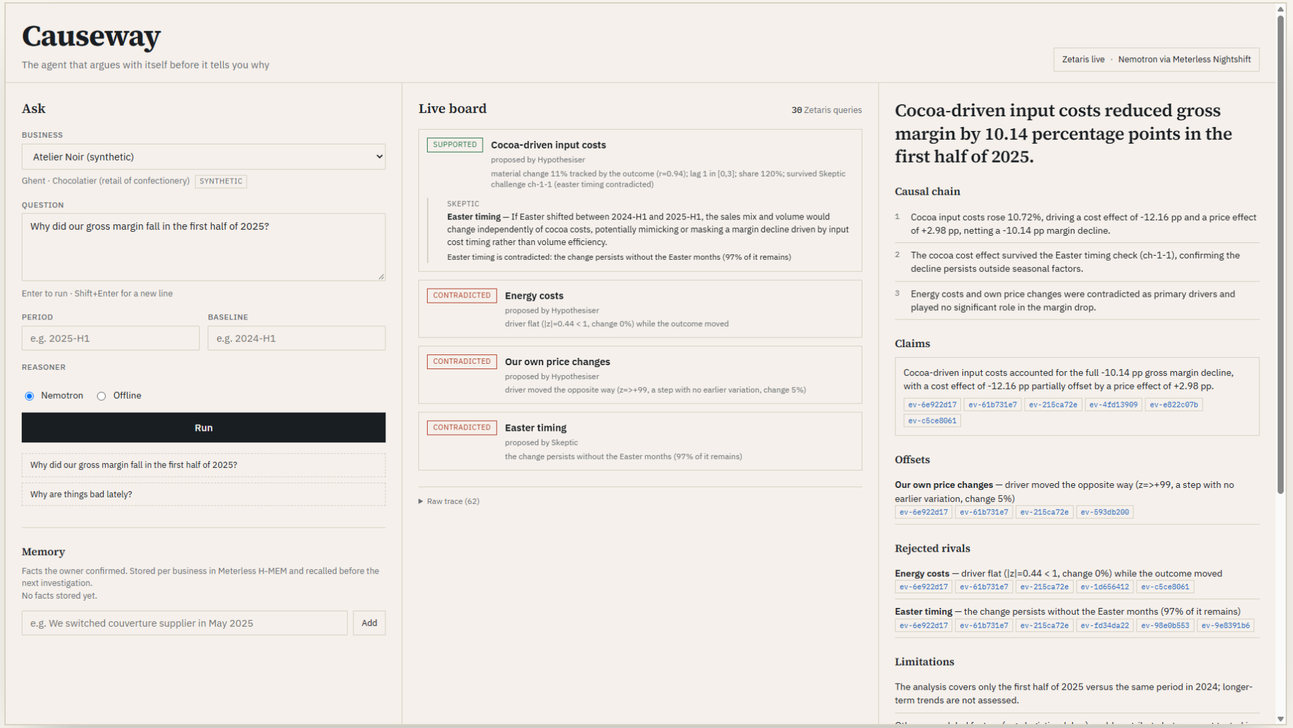
An investigation, not a paragraph.

Proposes rival causes, tests each one with SQL on the owner's ledger joined to live public data.

A Skeptic with veto power forces the confounder tests (Easter timing, own price changes).

Every claim on the answer card links to the query that produced its number.

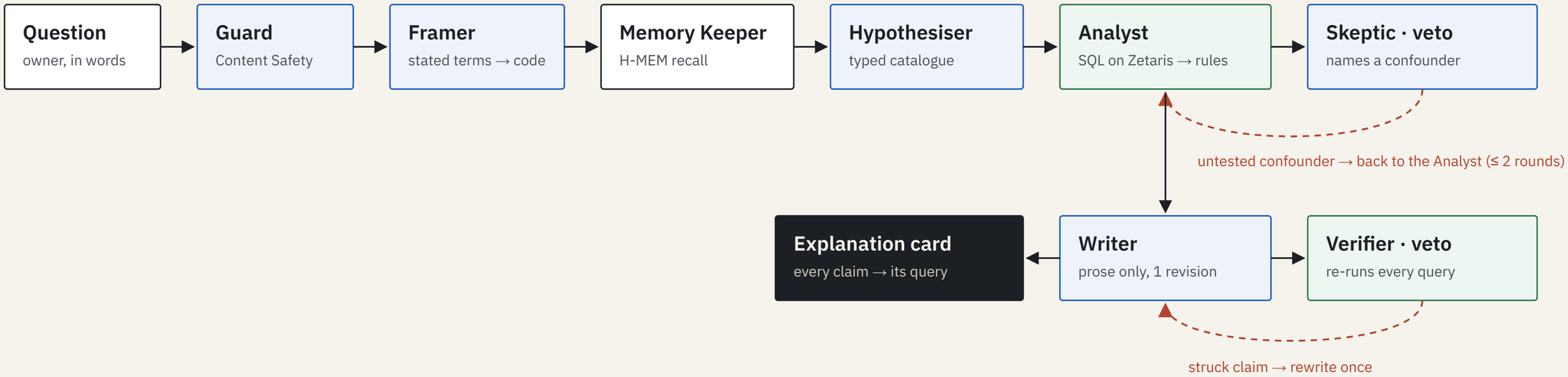
Owner memory changes the next answer.



Real run · Zetaris live · Nemotron via Meterless Nightshift · 30 Zetaris queries

AGENT WORKFLOW

Eight agents, one shared evidence graph. Models propose; code decides.



MODELS · METERLESS NIGHTSHIFT (PRIMARY PROVIDER) → NVIDIA NEMOTRON 3.5

Framer, Hypothesiser, Skeptic and Writer calls are routed by Nightshift; signed receipts record the routed model.
One scope per investigation, so the runtime's own memory and answer cache never cross runs. Failover: NVIDIA direct, then Nebius.

DATA · ZETARIS LIGHTNING

USL "causeway": ledger datasource (PostgreSQL)
+ 7 Eurostat REST tables, 5 tables, 4 foreign keys.
Only data path; a snapshot replays hash-identical.

STATE · METERLESS ENGINES

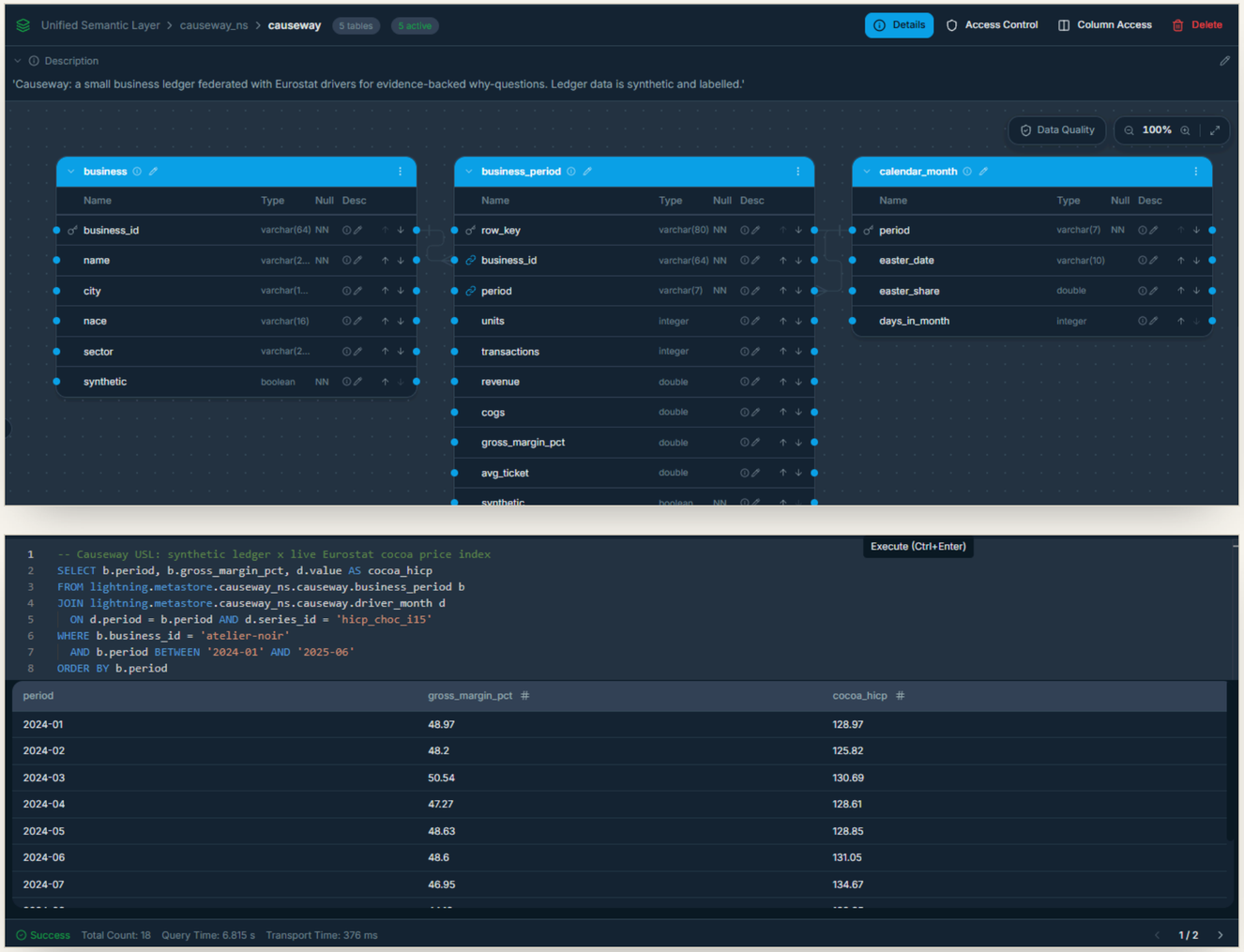
World Model: shared evidence graph for all agents.
H-MEM: owner facts + verdicts + trust ledger.
Markovian: bounded carryover between rounds.

Blue: model proposes · Green: code decides · Red: veto loops

EXPLAINABLE BY CONSTRUCTION

No model writes SQL or issues a verdict.

STEP	MODEL (NEMOTRON)	CODE
Frame	fills what the words leave open	a metric or period the question **states** is parsed by code and wins
Hypothesise	proposes rival causes	typed catalogue; out-of-catalogue proposals are rejected
Test	—	SQL on the Zetaris USL; z-scores, lagged association, contribution share; a **rule table** decides supported / contradicted
Challenge	Skeptic names a confounder	the confounder is tested; “nothing explains it” is challenged too
Explain	Writer drafts prose	Verifier re-executes queries (hash match); strikes numbers not in the evidence, wrong pp units, untested causes and numbers credited to the wrong cause



Real Zetaris UI · USL “causeway” · SQL Workspace: synthetic ledger × live Eurostat cocoa index

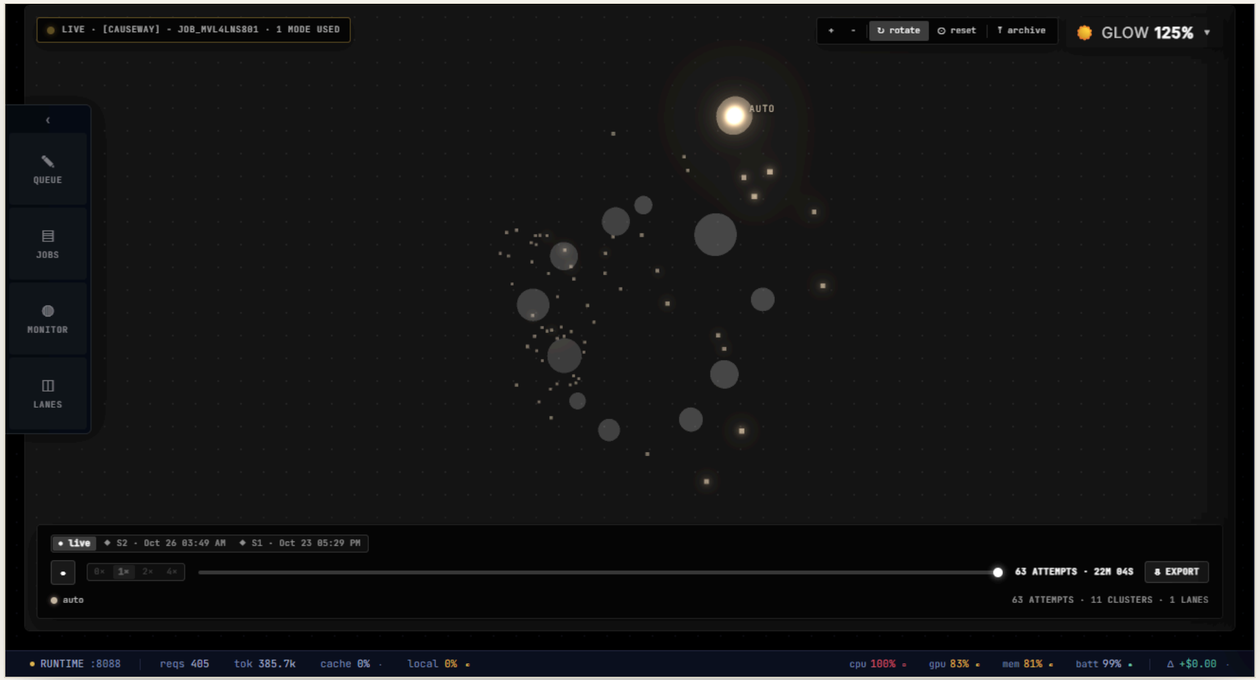
Ledger and public drivers in one semantic layer.

Federated: PostgreSQL ledger datasource + 7 Eurostat REST tables, unified in the USL `causeway` (5 tables, 4 foreign keys).

Every number has a query: the Analyst only runs SQL on the USL; each result is hashed and cited.

Replayable: a snapshot of the USL replays hash-identical (5/5 tables, 3/3 joins).

Live data moves: on 24 Oct Eurostat republished retail volume with an empty month; the parity check caught it, and one REST table was recreated.



Real Nightshift 1.2.3 window · live job “[Causeway]” · attempts and requests rise while Causeway runs

Nightshift is the primary provider.

Every reasoning call is routed by Nightshift to Nemotron 3.5, and receipts record the routed model: 316 calls across 7 Bench runs, all answered, 0 failovers.

One scope per investigation. Nightshift injected its own stored memories into our prompts; a per-run scope keeps every run independent.

H-MEM holds what the owner confirmed · **World Model** is the evidence graph · **Markovian** carries state between Skeptic rounds.

▼ Raw trace (62)

1	Orchestrator	start	{"input":{"question":"Why did our gross margin fall in …
2	ModelRouter	call	{"provider":"nvidia","model":"nvidia/nemotron-3.5-conte…
3	Guard	classify	{"safe":true,"source":"nemotron-content-safety","label"…
4	Memory Keeper	recall	{"facts":[]}
5	ModelRouter	call	{"provider":"meterless","model":"nvidia/nemotron-3.5-li…
6	Framer	model_call	{"provider":"meterless","model":"nvidia/nemotron-3.5-li…
7	Framer	frame	{"source":"nemotron","result":{"status":"ok","frame":{"…
8	Analyst	query_zetaris	{"hypothesis_id":"frame","evidence_id":"ev-6e922d17","p…
9	Analyst	query_zetaris	{"hypothesis_id":"frame","evidence_id":"ev-61b731e7","p…

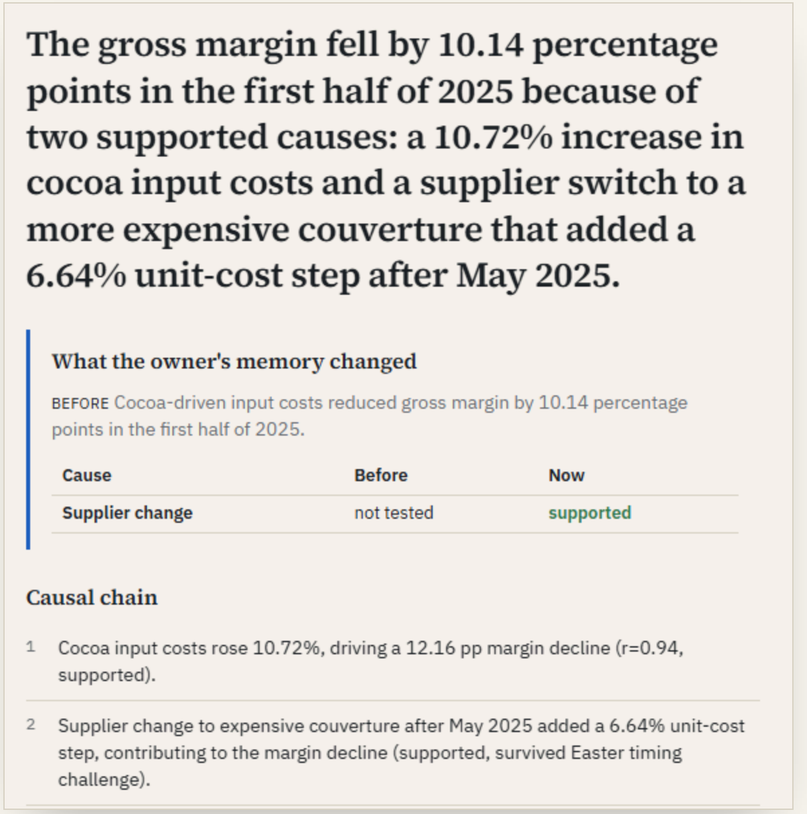
Causeway trace · guard → NVIDIA · reasoning → provider “meterless”

MEMORY CHANGES THE ANSWER

The owner adds one fact. The next answer tests it.

“We switched to a more expensive single-origin couverture supplier in May 2025.” It is stored in H-MEM, mapped to a testable cause with a start month, and recalled before the next investigation.





Real re-ask on Zetaris: supplier change goes from “not tested” to “supported” (unit-cost step z = 11.36), after surviving the Skeptic’s Easter challenge

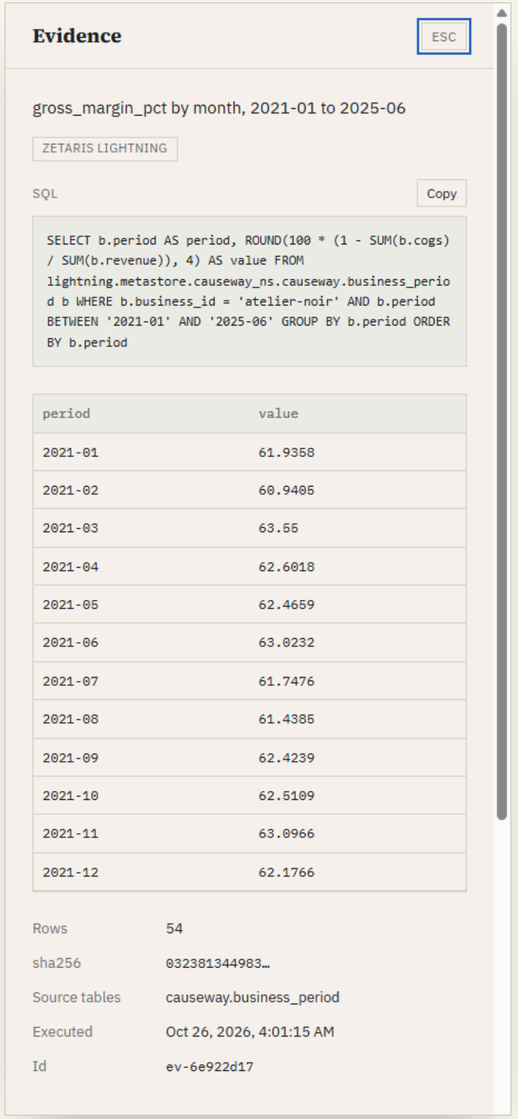
EVERY CLAIM HAS A RECEIPT

Click any claim: the SQL, the rows and the hash.

The Verifier **re-executes every cited query** and compares the result hash before the owner sees the card.

It **strikes** a claim whose number is in no evidence, whose “percentage points” are not pp in the evidence, that names a cause never tested, or that credits a number to the wrong cause.

A struck claim goes back to the Writer once; the headline verdict falls back to a code-built one.



CAUSEWAY BENCH · MEASURED, NOT CLAIMED

12 cases on 8 synthetic businesses with a hidden ground truth, driven by real Eurostat series.

12/12

main cause found

2/2

“nothing external explains it” controls passed

0

ungrounded numbers in the final prose

> **One false support, in every run (B11).** For a bike shop, Belgian retail bankruptcies rose 11 % and track the synthetic sales at r = 0.54, which passes a rule a shared trend can satisfy. It is a clean holdout case, so we report it instead of tuning the rule after seeing it.

Run through Meterless Nightshift, snapshot of the Zetaris USL. Tune/holdout split; holdout cases that ever prompted a fix are listed as no longer clean. Every run, including a miss (11/12 before a Framer fix), is in `bench/RUNS.md`.

Measures recovery of planted causes, not real-world causality.

Built to be run by someone else.

LAYER	CHOICE
Data	Zetaris Lightning (local Docker): USL, JDBC ledger datasource, REST tables; DuckDB snapshot replay
Models	NVIDIA Nemotron 3.5 Lightning, Super, Content Safety; Nebius failover
Runtime	Meterless Nightshift (provider contract v4), World Model, H-MEM, Markovian (vendored, unmodified, conformance 8/8, 13/13, 8/8)
App	TypeScript, Node 22, Fastify API + SSE, React UI, zod, 74 unit tests, Docker

Cursor

Headless Cursor sessions built the scaffold, the Meterless Runtime provider, the web UI and the README draft; every session is logged in the repo.

Run it

```
docker compose up
npm run ask -- --input input_examples/example_1.json
```

An empty .env runs the full loop on the bundled Zetaris snapshot with the offline reasoner. A fresh clone from GitHub passes the Docker test (16/16).

Unique

Adversarial and auditable: a Skeptic and a Verifier both hold vetoes; every number is a re-runnable Zetaris query.

Sponsor tech doing real work: Zetaris federates private and public data; Nightshift routes the reasoning; H-MEM turns an owner’s sentence into a test.

Honest evaluation: a public bench with its failures left in.

Next

Connect real accounting ledgers through Zetaris datasources, in place of the synthetic ones.

An accountant view across many client businesses.

More drivers (wages, rents, weather, local events) and a stricter association rule, measured on new holdout cases.

Monthly watch: re-run the question when Eurostat publishes.

Ask why. Get the evidence, not a guess.

Source, README and Bench: github.com/RuiCodex-dev/open-agent-causeway

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